Review for Unit Two

In this document, we will review the content from unit 2: Tidyverse and ggplot.

1 Cleaning data using tidyverse

To clean data, first a data file must be read in. We do this using read.csv or read_csv, which is the tidyverse equivalent. This whole example will be worked using the police brutality data set from class.

```
library(tidyverse)
dat.police = read_csv("Review/BrownIannuzi22.csv")
```

First, we use colnames() to get an idea of what columns are in the data. Then, we want to find the masked ip addresses that occur more than once, and rempve them. We will use tidyverse and pipes to do this. The code is commented to explain what each individual step is doing.

```
colnames(dat.police)
```

```
 [1] "masked.ip.id"   "age"            "gender"         "ethnicity"
 [5] "hispanic"       "SubjectId"      "Captcha1"       "Captcha2"
 [9] "Consent"        "UseData"        "BlackSuspect"   "RaceCheck"
[13] "GenderCheck"    "WeaponCheck"    "PrintCheck"     "WhiteSuspect"
[17] "RaceCheck_w"    "GenderCheck_w"  "WeaponCheck_w"  "PrintCheck_w"
[21] "Masculine_1"    "Masculine_2"    "Masculine_3"    "Masculine_4"
[25] "Masculine_5"    "Feminine_7"     "Feminine_8"     "Feminine_9"
[29] "Feminine_10"    "SuspectPain_1"  "SuspectPain_2"  "SuspectPain_3"
[33] "SuspectFault_1" "SuspectFault_2" "SuspectFault_3" "OfficerFault_1"
[37] "OfficerFault_2" "OfficerFault_3" "BeneSexism_1"   "BeneSexism_2"
[41] "BeneSexism_3"   "BeneSexism_4"   "BeneSexism_5"   "HostSexism_1"
[45] "HostSexism_2"   "HostSexism_3"   "HostSexism_4"   "HostSexism_5"
[49] "HostSexism_6"
```

```
repeats = table(dat.police$masked.ip.id) > 1 # creates a proportion table, where each row keeps track of th
# setting the values to > 1 turns the table into true/false, where true is the repeated ids
not.repeats = names(table(dat.police$masked.ip.id))[-which(repeats)] # gives the not repeated ids
dat.police.cleaned = dat.police |>
    # Group by ip address
group_by(masked.ip.id) |>
    #Keep observations that occur once
filter(n() == 1) |>
    #Ungroup data
ungroup() |>
    # Keep those passing captcha 1
filter(Captcha1 == 1 ) |>
    # Keep those passing captcha 2
filter(trimws(tolower(Captcha2)) == "cap") |>
    # Keep those who consented
filter(Consent == 1) |>
    # Change treatment into one column and remove those without a treatment
mutate(Treatment = case_when(!is.na(WhiteSuspect) ~ "White Suspect", !is.na(BlackSuspect) ~ "Black Suspect"
```

```r
filter(!is.na(Treatment)) |>
    # Keep those who passed their race attention check
filter( (!is.na(RaceCheck) & (RaceCheck == 1) |
    (!is.na(RaceCheck_w) & (RaceCheck_w == 1)))
) |>
    # Keep those who said it was ok to use their data or said nothing
filter(UseData == 1 | is.na(UseData)) |>
    # Remove those who did not respond to primary questions
filter(!(SubjectId %in% c(7603, 6562, 1806)))

dim(dat.police.cleaned)
```

```
[1] 466  50
```

This chunk of code below combines the variables which are on scales into one variable, and takes the mean of that variable for each individual observation. For example, the first row of the dataset has Masculine = 73, which is the average of the answers to maculitnity questions for that responder. The same process is used for the rest of the variables.

```r
dat.police.cleaned2 <- dat.police.cleaned %>%
 mutate(Feminine = . |> dplyr::select(starts_with("Feminine")) |> rowMeans(na.rm=T)) %>%
 mutate(Masculine = . |> dplyr::select(starts_with("Masculine")) |> rowMeans(na.rm=T)) %>%
 mutate(SuspectPain = . |> dplyr::select(starts_with("SuspectPain")) |> rowMeans(na.rm=T)) %>%
 mutate(SuspectFault = . |> dplyr::select(starts_with("SuspectFault")) |> rowMeans(na.rm=T)) %>%
 mutate(OfficerFault = . |> dplyr::select(starts_with("OfficerFault")) |> rowMeans(na.rm=T)) %>%
 mutate(BeneSexism = . |> dplyr::select(starts_with("BeneSexism")) |> rowMeans(na.rm=T)) %>%
 mutate(HostSexism = . |> dplyr::select(starts_with("HostSexism")) |> rowMeans(na.rm=T)) |>
 dplyr::select(Treatment, Masculine, Feminine,
 SuspectPain, SuspectFault, OfficerFault,
 BeneSexism, HostSexism)
```

2 Summarizing data

There are two things that that should always be made when summarizing data: A plot, and a numerical summary. What kind of plot and what numerical summaries depends on the type of variables in the data: categorical or quantitative.

Categorical variables are not generally made of up numbers, and observations are a part of one of several groups or categories. There are two types of categorical variables: Nominal, which assumes groups are made of up named categories, and Ordinal, which assumes groups are made up of ordered categories. Examples of a nominal categorical variable include maritial status or car brand. An example of an ordinal categorical variable is satisfaction rankings on a survey: very bad, bad, ok, good, very good.

When there is one categorical variable, two graphical options are the bar chart and the doughnut plot. For a numerical summary, we want to calculate the sample proportion of observations that belong to each level of the categorical variable. This corresponds to the equation $p = \frac{x}{n}$, where x is the number of observations in the desired group, and n is the number of total observations.

Here is an example of a numerical summary:

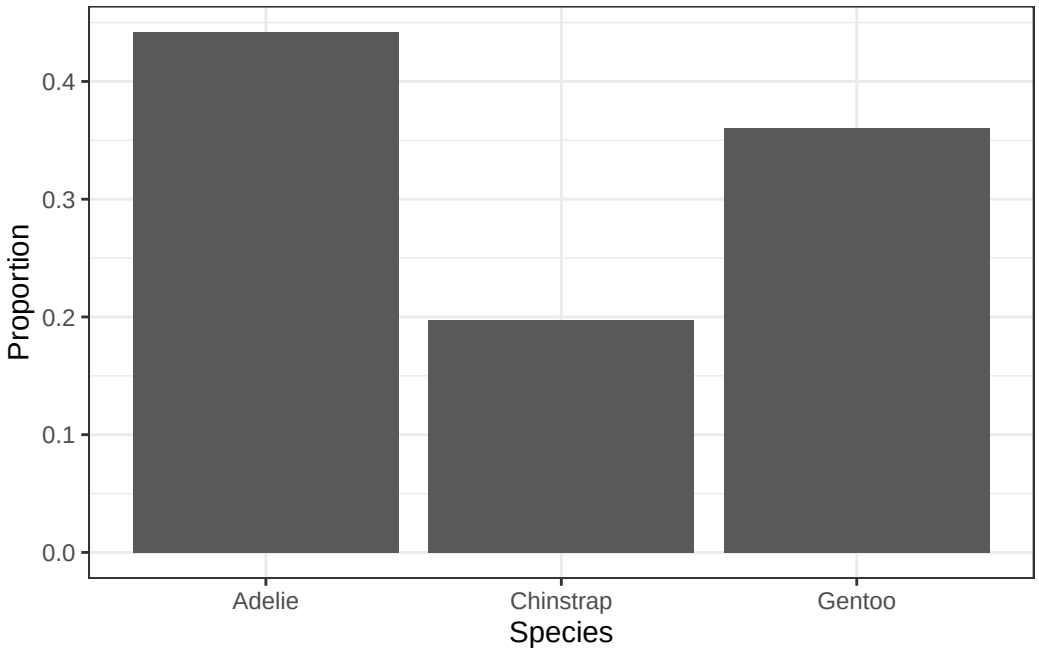
```r
penguins |>
    group_by(species) |>
    summarize(n = n()) |>
    mutate(proportion = n/sum(n))
```

```
# A tibble: 3 x 3
  species       n proportion
  <fct>     <int>      <dbl>
1 Adelie      152      0.442
```

```
2 Chinstrap    68      0.198
3 Gentoo      124      0.360
```

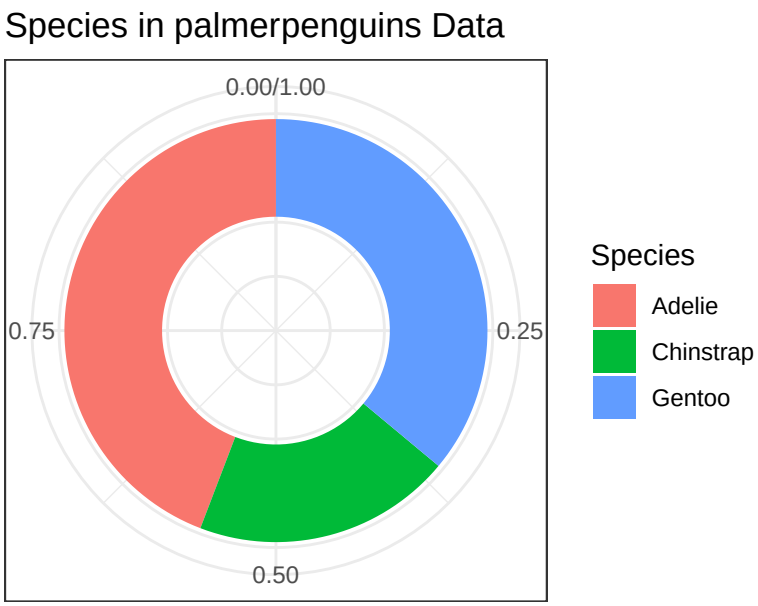
Here is an example of a bar plot.

```r
ggplot(penguins) +
    geom_bar(aes(x = species, y = after_stat(count/sum(count)))) +
    theme_bw() +
    labs(x = "Species", y = "Proportion")
```



And here is an example of a doughnut plot.

```r
ggplot(penguins) +
    geom_bar(aes(x = 2,
    y = after_stat(count/sum(count)), # Specify y-axis
    fill = species)) + # Specify fill coloring
    coord_polar("y", start=0) + # Specify a pie chart
    xlim(0.5, 2.5) + # To get a doughnut plot
    theme_bw() + # A cleaner theme
    labs(x = "", # Specify axis labels
    y = "",
    fill = "Species") + # Specify legend labels
    ggtitle("Species in palmerpenguins Data") + # Plot title
    theme(axis.text.y = element_blank(), # Removes y-axis ticks
    axis.ticks.y = element_blank())
```



For two categorical variables, we employ the same strategy for a numerical summary as we did with one, with the addition of a new variable. This is called a contingency table.

```
penguins |> # Data to start with
    group_by(island, species) |> # Group by island and species
    summarize(n = n()) |> # Count observations
    group_by(island) |> # Group by island
    mutate(prop = n / sum(n))
```

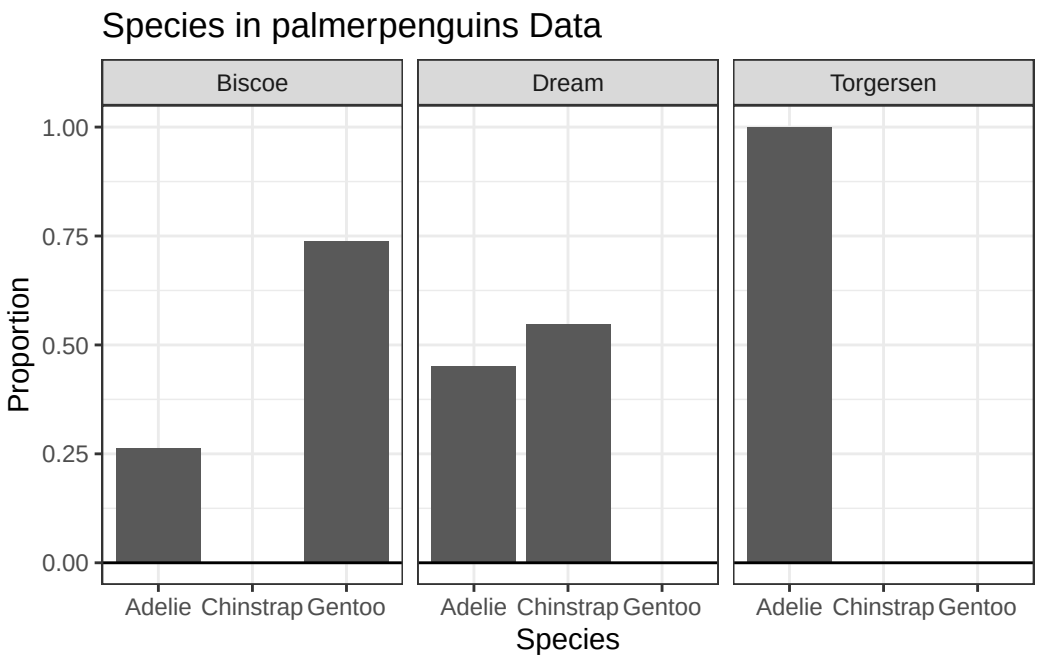
```
# A tibble: 5 x 4
# Groups:   island [3]
  island    species       n  prop
  <fct>     <fct>     <int> <dbl>
1 Biscoe    Adelie       44 0.262
2 Biscoe    Gentoo      124 0.738
3 Dream     Adelie       56 0.452
4 Dream     Chinstrap    68 0.548
5 Torgersen Adelie       52 1
```

```
conditional.dat <- penguins |> # Data to start with
count(island, species) |> # Count observations
mutate(prop = n / sum(n), .by = island)
```

To graph two categorical variables, we can employ grouped doughnut plots, a mosaic plot, or grouped bar plots. The code for these is below.

Grouped bar plots:

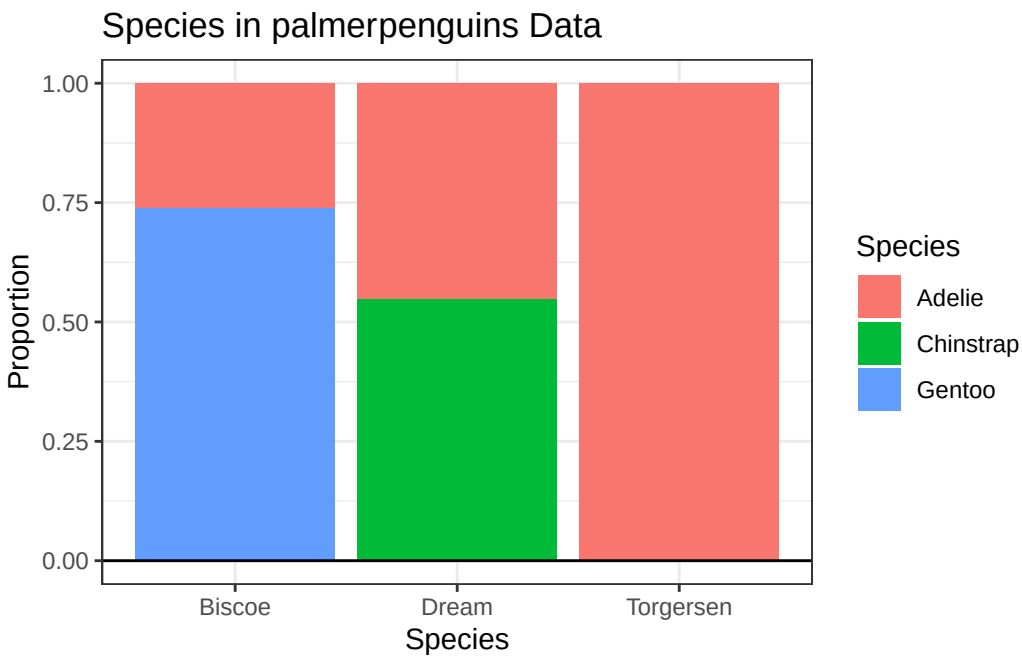
```
ggplot(conditional.dat) + # Specify Data
    geom_bar(aes(x = species, # Specify x-axis
    y = prop), # Specify y-axis
    stat="identity") + # We gave the statistic
    geom_hline(yintercept = 0)+ # Plot x-axis
    theme_bw() + # A cleaner theme
    labs(x = "Species", # Specify axis labels
    y = "Proportion") +
    ggtitle("Species in palmerpenguins Data") + # Plot title
    facet_wrap(~island)
```



Note that this is nearly the same code as the singular bar plot, except prop is now in the data, and at the end we use facet_wrap() to separate by island.

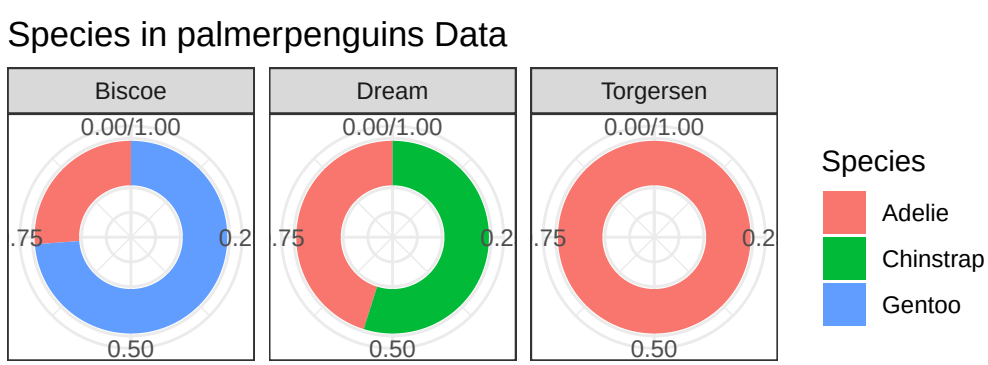
This is another method of making a grouped bar plot. The difference is that it is all on one graph, and the color of the bars changes according to the species.

```
    ggplot(conditional.dat) +
    geom_bar(aes(x = island, # Specify x-axis
    y = prop, # Specify y-axis
    fill = species), # Specify fill color
    stat="identity") + # We gave the statistic
    geom_hline(yintercept = 0)+ # Plot x-axis
    theme_bw() + # A cleaner theme
    labs(x = "Species", # Specify axis labels
    y = "Proportion",
    fill = "Species") + # Specify legend label
    ggtitle("Species in palmerpenguins Data")
```



The grouped doughnut plot is largely the same as the singular doughnut plot, simply with more plots.

```
ggplot(conditional.dat) + # Specify Data
geom_bar(aes(x = 2, # Specify x-axis
y = prop, # Specify y-axis
fill = species), # Specify fill coloring
stat = "identity") +
coord_polar("y", start=0) + # Specify a pie chart
xlim(0.5, 2.5) + # To get a doughnut plot
theme_bw() + # A cleaner theme
labs(x = "", # Specify axis labels
y = "",
fill = "Species") + # Specify legend labels
ggtitle("Species in palmerpenguins Data") + # Plot title
theme(axis.text.y = element_blank(), # Removes y-axis ticks
axis.ticks.y = element_blank()) +
facet_wrap(~ island)
```



One continuous variable can be plotted with a histogram, boxplot, or violin plot. In a numerical summary, we want the mean, median, standard deviation, IQR, skewness, and kurtosis.

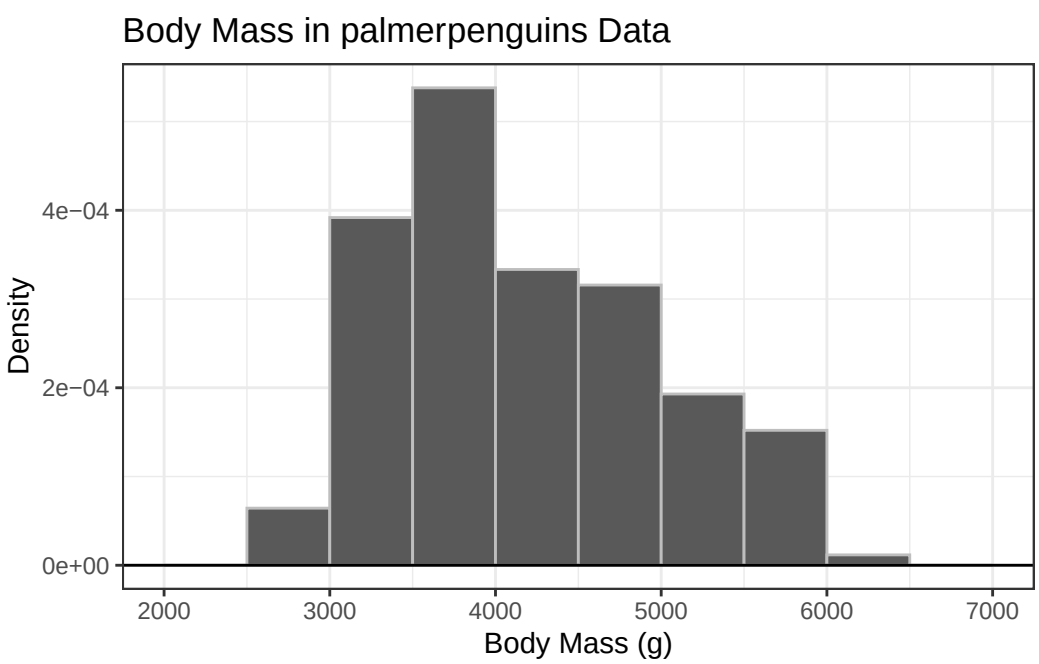
Here is the code for a numerical summary. Note it has the same basic form as the categorical variables, there are just different things being calculated in the summarize() command. In this case, the continuous variable is the body mass.

```r
penguins |>
    drop_na(body_mass) |>
    summarize(
    Mean = mean(body_mass, na.rm = T),
    SD = sd(body_mass, na.rm = T),
    Median = median(body_mass, na.rm = T),
    IQR = IQR(body_mass, na.rm = T),
    Skewness = e1071::skewness(body_mass, na.rm = T),
    "Excess Kurtosis" = e1071::kurtosis(body_mass, na.rm = T)
)
```

```
      Mean       SD Median  IQR  Skewness Excess Kurtosis
1 4201.754 801.9545   4050 1200 0.4662117        -0.73952
```

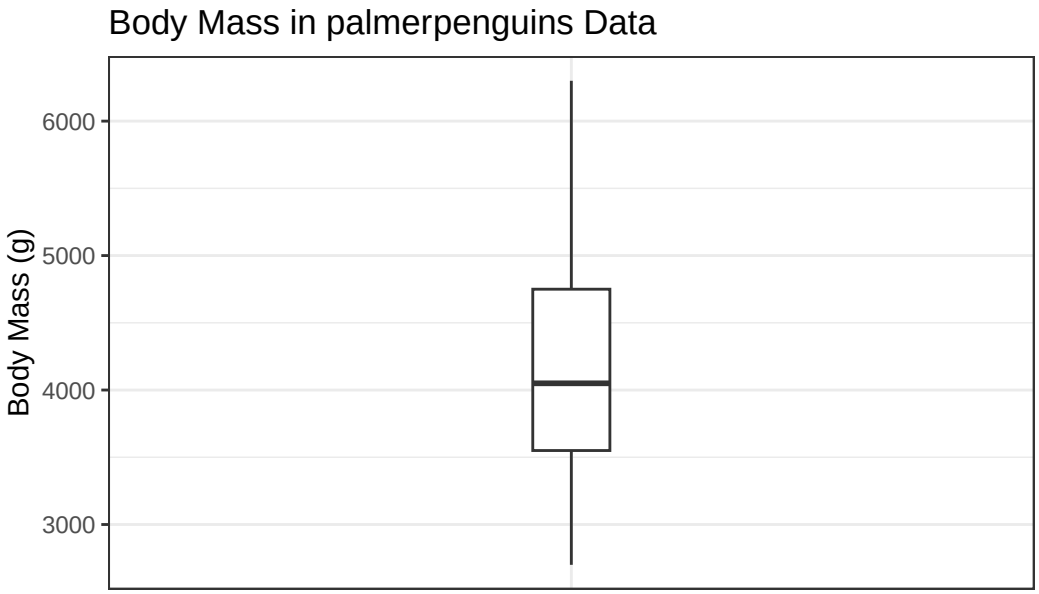
Here is the code for a histogram:

```r
ggplot(penguins) + # Specify Data
    geom_histogram(aes(x = body_mass, # Specify x-axis
    y = after_stat(density)), # Specify y-axis
    breaks = seq(2000, 7000, 500), # Specify bins
    color = "gray")+
    geom_hline(yintercept = 0)+ # Plot x-axis
    theme_bw() + # A cleaner theme
    labs(x = "Body Mass (g)", # Specify axis labels
    y = "Density") +
    ggtitle("Body Mass in palmerpenguins Data")
```



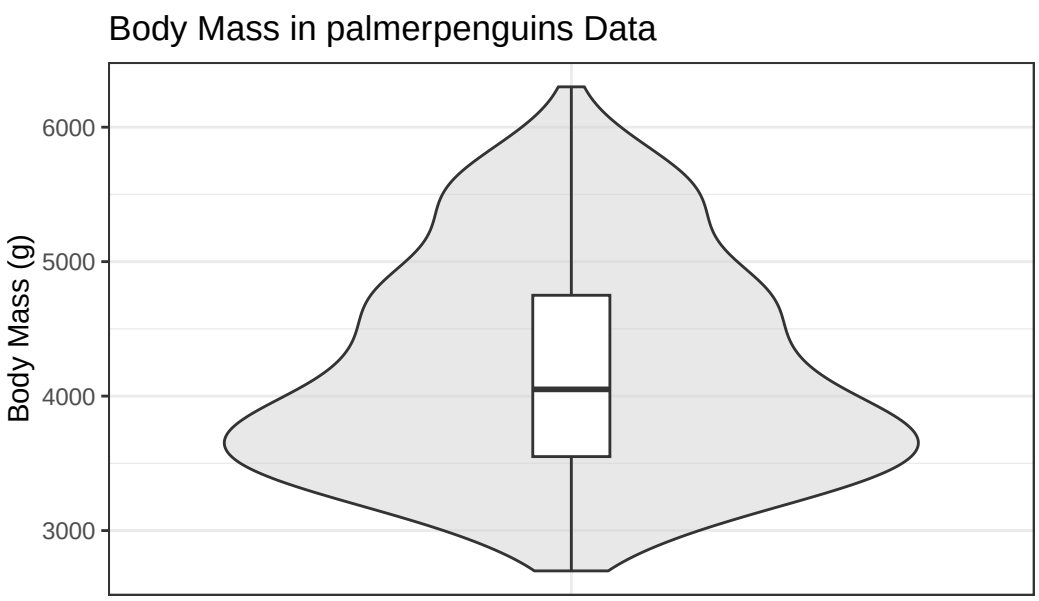
Here is the code for a box plot:

```r
ggplot(penguins) + # Specify Data
    geom_boxplot(aes(x = "", # Specify no grouping
    y = body_mass), # Specify y-axis
    width = 0.1)+
    theme_bw() + # A cleaner theme
    labs(x = "", # Specify axis labels
    y = "Body Mass (g)") +
    ggtitle("Body Mass in palmerpenguins Data") + # Plot title
    theme(axis.ticks.x = element_blank())
```



Here is the code for a violin plot. Note the similarity to the boxplot, just with the addition of the geom_violin() layer.

```
ggplot(penguins) + # Specify Data
geom_violin(aes(x = "", # Specify no grouping
y = body_mass), # Specify y-axis
fill = "lightgrey", alpha=0.5) + # A transparent light grey
geom_boxplot(aes(x = "", # Specify no grouping
y = body_mass), # Specify y-axis
width = 0.1)+
theme_bw() + # A cleaner theme
labs(x = "", # Specify axis labels
y = "Body Mass (g)") +
ggtitle("Body Mass in palmerpenguins Data") + # Plot title
theme(axis.ticks.x = element_blank())
```



For one continuous and one categorical variable, we take the plots used for a categorical variable and group them. This gives us grouped histograms, grouped boxplots, and grouped violin plots. We use a similar process for the numerical summary, calculating group means, group medians, group IQR, and group standard deviation.

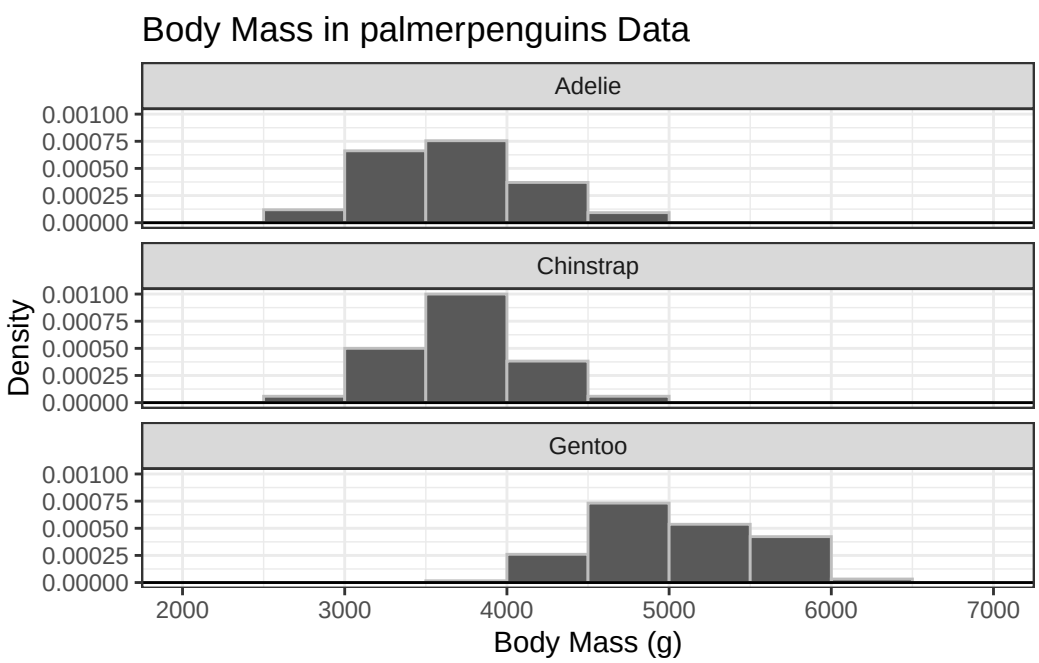
Here is the code for a numerical summary. Note the similarity to the numerical summary for one categorical variable - the only difference is the group_by at the beginning.

```
penguins |>
group_by(species) |>
summarize(
Mean = mean(body_mass, na.rm = T),
SD = sd(body_mass, na.rm = T),
Median = median(body_mass, na.rm = T),
IQR = IQR(body_mass, na.rm = T),
Skewness = e1071::skewness(body_mass, na.rm = T),
"Excess Kurtosis" = e1071::kurtosis(body_mass, na.rm = T)
)
```

```
# A tibble: 3 x 7
  species    Mean    SD Median   IQR Skewness `Excess Kurtosis`
  <fct>     <dbl> <dbl>  <dbl> <dbl>    <dbl>             <dbl>
1 Adelie    3701.  459.   3700  650    0.280             -0.626
2 Chinstrap 3733.  384.   3700  462.   0.237              0.363
3 Gentoo    5076.  504.   5000  800    0.0679            -0.779
```

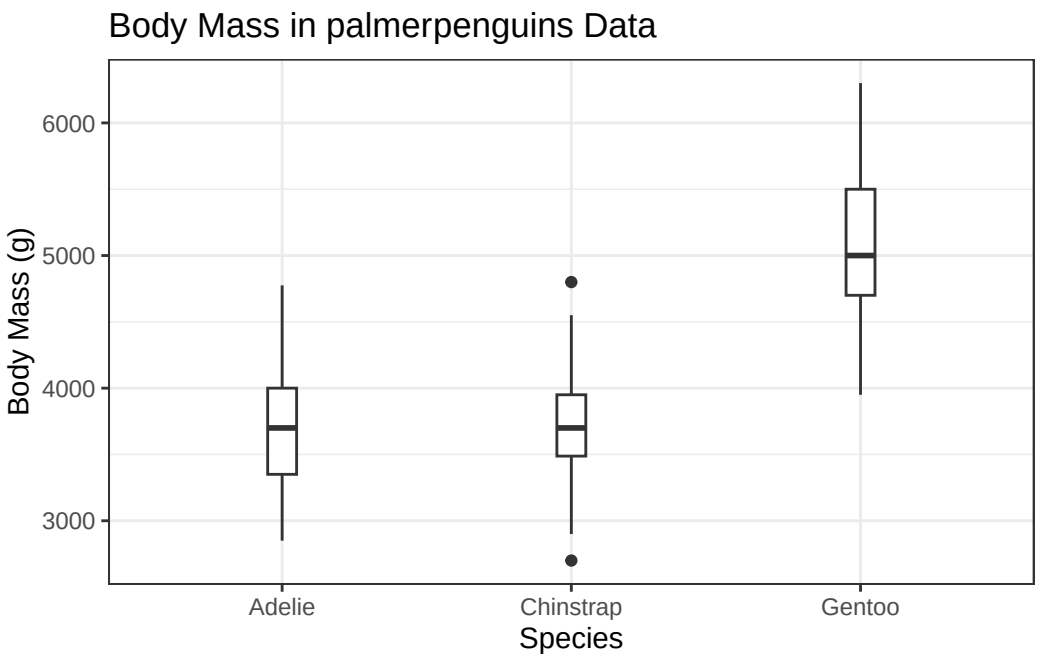
For the grouped histogram, we take the code for the histogram and add facet_wrap() at the end to group it by species.

```
ggplot(penguins) + # Specify Data
    geom_histogram(aes(x = body_mass, # Specify x-axis
    y = after_stat(density)), # Specify y-axis
    breaks = seq(2000, 7000, 500), # Specify bins
    color = "gray")+
    geom_hline(yintercept = 0)+ # Plot x-axis
    theme_bw() + # A cleaner theme
    labs(x = "Body Mass (g)", # Specify axis labels
    y = "Density") +
    ggtitle("Body Mass in palmerpenguins Data") + # Plot title
    facet_wrap(~species, nrow =3)
```



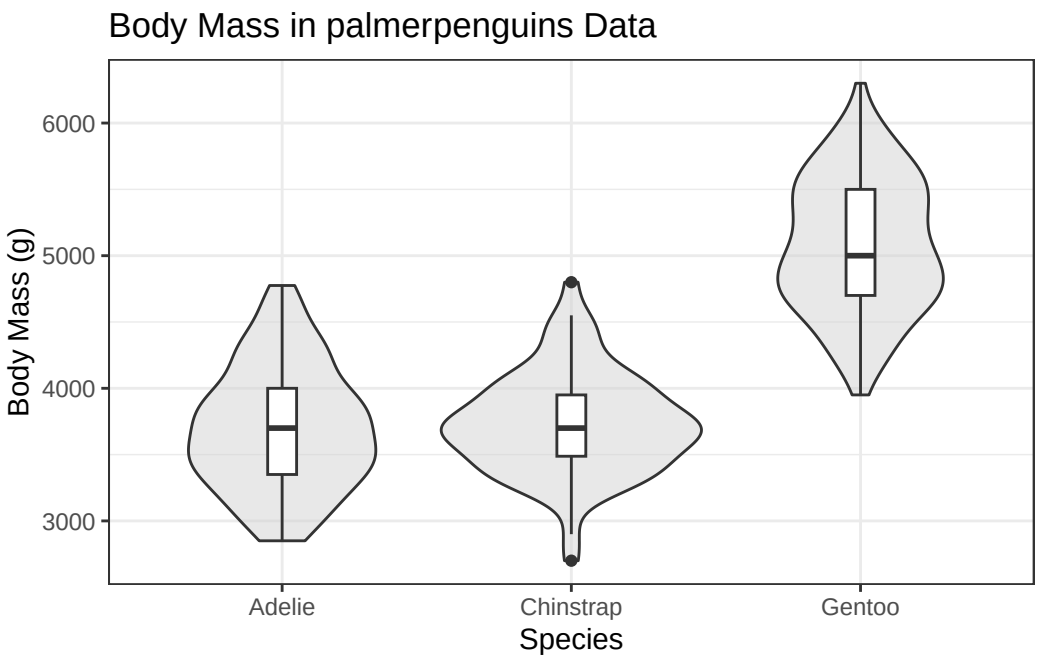
Here is the code for a grouped boxplot.

```r
ggplot(penguins) + # Specify Data
    geom_boxplot(aes(x = species, # Specify no grouping
    y = body_mass), # Specify y-axis
    width = 0.1)+
    theme_bw() + # A cleaner theme
    labs(x = "Species", # Specify axis labels
    y = "Body Mass (g)") +
    ggtitle("Body Mass in palmerpenguins Data")
```



Here is the code for a grouped violin plot.

```r
ggplot(penguins) + # Specify Data
geom_violin(aes(x = species, # Specify no grouping
y = body_mass), # Specify y-axis
fill = "lightgrey", alpha=0.5) + # A transparent light grey
geom_boxplot(aes(x = species, # Specify no grouping
y = body_mass), # Specify y-axis
width = 0.1)+
theme_bw() + # A cleaner theme
labs(x = "Species", # Specify axis labels
y = "Body Mass (g)") +
ggtitle("Body Mass in palmerpenguins Data")
```

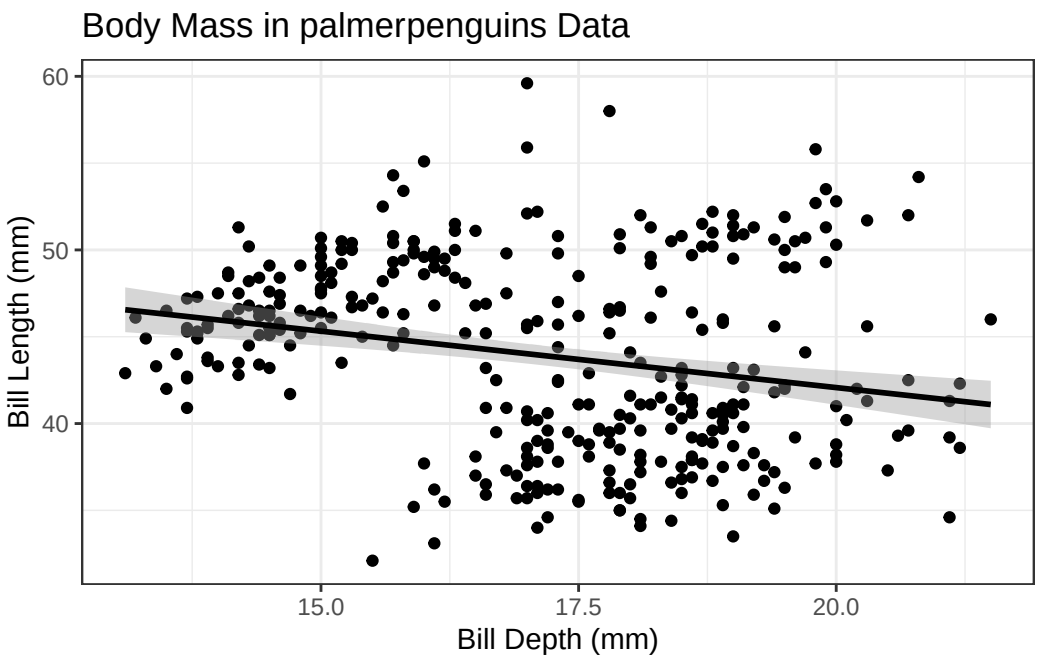


For two continuous variables, we only make the scatter plot for a graph. For the numerical summary, we calculate Pearson and Spearman's Correlations, and Kendall's Tau. The code for these is below.

```
penguins |>
drop_na(bill_dep, bill_len) |>
summarize(
Pearson = cor(bill_dep, bill_len, method="pearson"),
Spearman = cor(bill_dep, bill_len, method="spearman"),
Kendall = cor(bill_dep, bill_len, method="kendall")
)
```

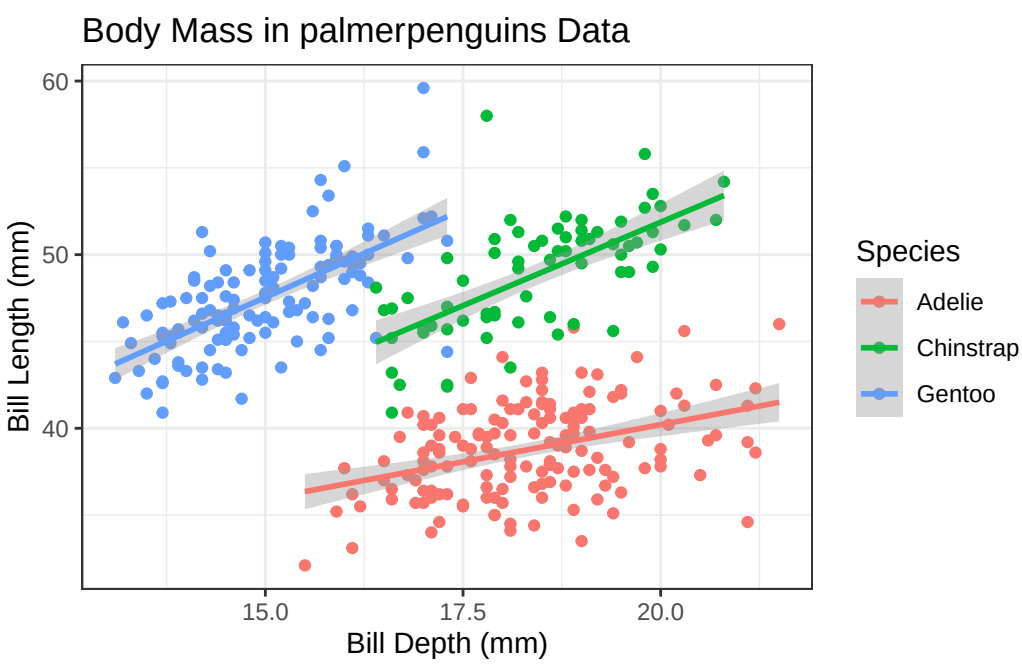
```
     Pearson   Spearman    Kendall
1 -0.2350529 -0.2217492 -0.1228502
```

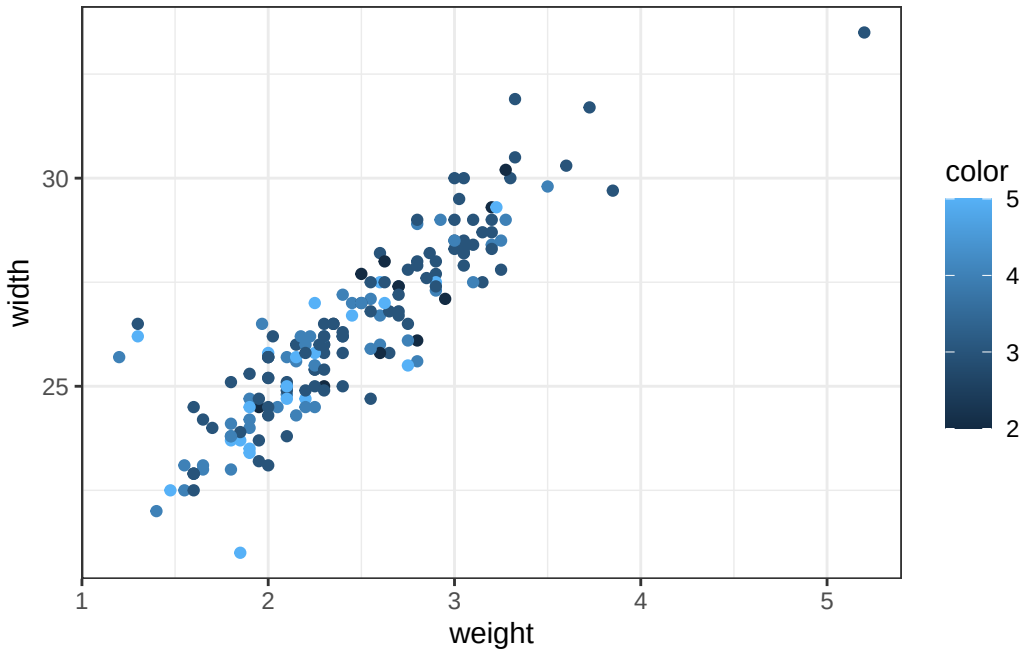
```
ggplot(penguins) + # Specify Data
geom_point(aes(x = bill_dep, # Specify x-axis
y = bill_len)) + # Specify y-axis
geom_smooth(aes(x = bill_dep, # Specify x-axis
y = bill_len), # Specify y-axis
method = "lm", # Specify best fit line
color = "black") + # Specify line color
theme_bw() + # A cleaner theme
labs(x = "Bill Depth (mm)", # Specify axis labels
y = "Bill Length (mm)") +
ggtitle("Body Mass in palmerpenguins Data") #
```



We can expand upon the scatterplot to group it by a categorical variable.

```
ggplot(penguins) + # Specify Data
geom_point(aes(x = bill_dep, # Specify x-axis
y = bill_len, # Specify y-axis
color = species)) + # Specify color
geom_smooth(aes(x = bill_dep, # Specify x-axis
y = bill_len, # Specify y-axis
color = species), # Specify color
method = "lm") + # Specify best fit line
theme_bw() + # A cleaner theme
labs(x = "Bill Depth (mm)", # Specify axis labels
y = "Bill Length (mm)",
color = "Species") + # Specify legend labels
ggtitle("Body Mass in palmerpenguins Data")
```



Here I practice with a different dataset: Horseshoe crabs. I chose two continuous variables: weight and width, and one categorical: color. There appears to be a positive correlation between width and weight, but there is not seemingly any correlation between color and width, or color and weight.

```
dat = read_csv("Review/Brockmann96.csv")

ggplot(dat) +
    geom_point(aes(x = weight, y = width, color = color)) +
    theme_bw()
```